



## AMP1006000-FUNC-100W

### Product Features

- Frequency range from 100 ~ 6000MHz for 10-300M bandwidth
- Class AB GaN design
- Instantaneous wide bandwidth
- 50 Ohm Input/Output impedance
- Small size and Light weight
- High reliability and Ruggedness
- Ultra-wide digital interference source

### Description

The AMP1006000-100W-28V is designed for general purpose. Operating frequency range is from 100~6000MHz,Customize any 10-300 MHz bandwidth.Gallium Nitride on GaN Technology is used and attached on a copper sub carrier.Improved thermal handling by patented technology.

### Electrical Specifications @ $V_{DC}=24-32V$ (Wide voltage input) , $T_C=50^{\circ}C$ , 50 $\Omega$ System

| PARAMETER                                | UNIT | MIN | TYP  | MAX  | SYMBOL               |
|------------------------------------------|------|-----|------|------|----------------------|
| Operating Frequency                      | MHz  | 100 | -    | 6000 | F <sub>O</sub>       |
| Operating Bandwidth                      | MHz  | -   | 300  | -    | BW                   |
| Peak Output Power<br>@CW Signal          | W    | -   | 100  | -    | P <sub>SAT</sub>     |
| Input Power                              | dBm  | -   | -    | -    | P <sub>IN</sub>      |
| Maximum Input Power<br>@Without Damage   | dBm  |     | -    | -    | P <sub>IN, MAX</sub> |
| Power Gain<br>@Input Power 0dBm          | dB   | -   | 50±2 | -    | G <sub>P</sub>       |
| Gain Flatness<br>@Input Power 0dBm       | dB   | -   | -    | ±1   | ΔG <sub>P</sub>      |
| Efficiency                               | %    | 50  | 52   | 55   | E <sub>FF</sub>      |
| Harmonics 1 to N<br>@Output Power100W    | dBc  | 12  | 13   | 15   | H <sub>N</sub>       |
| Spurious Level                           | dBc  | 60  | -    | -    | Spur                 |
| Input VSWR                               | dB   | -   | -    | -    | VSWR                 |
| Output VSWR                              | dB   | -   | -    | 2:1  | VSWR                 |
| Operating Voltage                        | V    | 24  | 28   | 32   | V <sub>DC</sub>      |
| Current Consumption<br>@Output Power100W | A    | -   | 7.0  | -    | I <sub>DC</sub>      |

### Environmental Specifications

| PARAMETER                         | UNIT | RATING    | SYMBOL           |
|-----------------------------------|------|-----------|------------------|
| Operating Case Temperature        | °C   | -20 ~ 85  | T <sub>C</sub>   |
| Storage Temperature               | °C   | -40 ~ 105 | T <sub>STG</sub> |
| Relative Humidity(Non-condensing) | RH   | 95        | %                |

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## AMP4006000-FUNC-100W

### Operating Voltages

| PARAMETER               | UNIT | NOMINAL VOLTAGE | VOLTAGE ACCURACY | SYMBOL          |
|-------------------------|------|-----------------|------------------|-----------------|
| Operating Voltage       | V    | 24-32           | ± 2%             | V <sub>DC</sub> |
| HPA Enable Voltage      | V    | 5-32            |                  | V <sub>DC</sub> |
| Current Monitor Voltage | V    | -               |                  |                 |
| Temp Monitor Voltage    | V    | -               |                  |                 |

### Mechanical Specifications

| PARAMETER    | UNIT | TYP                                    |
|--------------|------|----------------------------------------|
| Mass         | kg   | 0.8                                    |
| Dimension    | mm   | 170 x94 x 26.5<br>(Without Connectors) |
| RF Connector | -    | MCX Female : RF Input                  |
|              |      | N Female : RF Output                   |
| DC Connector | -    | Power Line, Male : Supply              |
| Cooling      |      | External Heat-sink Required            |

### DC INTERFACE CONNECTOR – [Hybrid D-Sub 6-Pin, Male]

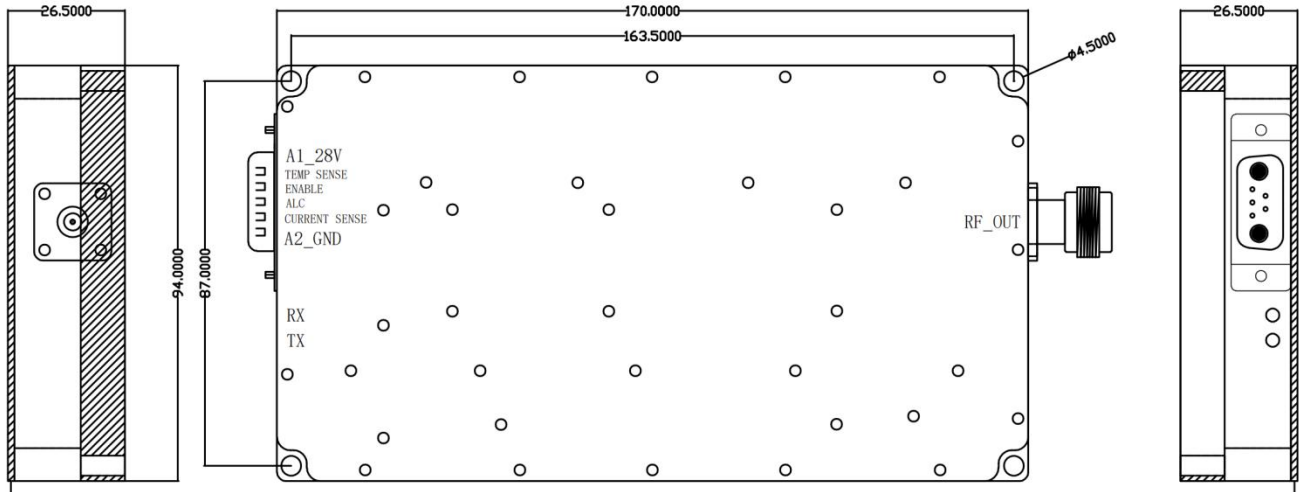
| Pin# | Description | Specifications                                          |
|------|-------------|---------------------------------------------------------|
| 1    | A1_28V      | +28VDC                                                  |
| 2    | TEMP        | Analog voltage relative to Module's Temperature@10mV/°C |
| 3    | Enable      | 3.3-5V                                                  |
| 4    | ALC         | Automatic Level Control                                 |
| 2    | CURR        | Analog voltage relative to IDD@100mV per Ampere         |
| 6    | A2_GND      | Ground                                                  |



## AMP4006000-FUNC-100W

### Outline Drawing

\* Unit: mm[inch] | Tolerance  $\pm 0.2$ [.0078]



100-6000M

### Side View [3D]

